4.8.3

$p=1$, $k$, $\sigma_1^2 \neq \sigma_2^2 \neq \dots \sigma_k^2$

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)$$

Bayes Theorem tells us:

$$p_k(x) = \frac{\pi_k f_k(x)}{\sum_{l}^{k} \pi_l f_l(x)}$$

$$\Rightarrow p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)}{\sum_{l=1}^{k} \pi_l \frac{1}{\sqrt{2\pi}\sigma_l} \exp\left(-\frac{(x-\mu_l)^2}{2\sigma_l^2}\right)}$$

$$\Rightarrow \log p_k(x) = \log\left(\frac{\pi_k f_k(x)}{\sum_{l=1}^{k} \pi_l f_l(x)}\right)$$

$$\Rightarrow \log p_k(x) = \log(\pi_k f_k(x)) - \log\left(\sum_{l=1}^{k} \pi_l f_l(x)\right)$$

Since $\log\left[\sum_{k=1}^{K} \pi_k f_k(x)\right]$ does not dependent on class index $k$, we can ignore it

$$\Rightarrow \delta_k(x) = \log(\pi_k f_k(x))$$

$$\Rightarrow \delta_k(x) = \log\left[\pi_k \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)\right]$$

$$\Rightarrow \delta_k(x) = \log(\pi_k) + \log\left(\frac{1}{\sqrt{2\pi}\sigma_k}\right) - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

$$= \log(\pi_k) - \frac{1}{2}\log(2\pi) - \log(\sigma_k) - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

Here we can drop $-\frac{1}{2}\log(2\pi)$

$$\Rightarrow \delta_k(x) = -\frac{(x-\mu_k)^2}{2\sigma_k^2} - \log(\sigma_k) + \log(\pi_k)$$

$$= -\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} - \log(\sigma_k) + \log(\pi_k)$$

Compared to LDA in 4.18, we can see that Here QDA has extra $-\frac{x^2}{2\sigma_k^2}$ term, which is quadratic in x.

Unlike LDA, QDA does not assume $\sigma_1^2 = \sigma_2^2 = \sigma_k^2$, so the quadratic x term cannot be canceled when comparing discriminant functions. Thus QDA Bayes classfier is quadratic in X.

4.6.7.

$$P(Y=\text{yes} \mid X=4) = \frac{P(Y=\text{yes}) f_{\text{yes}}(4)}{P(Y=\text{yes}) f_{\text{yes}}(4) + P(Y=\text{no}) f_{\text{no}}(4)}$$

$$= \frac{0.8 f_{\text{yes}}(4)}{0.8 f_{\text{yes}}(4) + 0.2 f_{\text{no}}(4)}$$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

for Yes, we use $\mu=10$, $\sigma^2=36$, $x=4$

$$\Rightarrow f_{\text{yes}}(4) = \frac{1}{\sqrt{72\pi}} e^{-0.5} \approx 0.04033$$

For no, we use $\mu=0$, $\sigma^2=36$, $x=4$

$$f_{\text{no}}(4) = \frac{1}{\sqrt{72\pi}} e^{-2/9} \approx 0.05324.$$

$$P(Y=\text{Yes} \mid X=4) = \frac{0.8 \times 0.04033}{0.8 \times 0.04033 + 0.2(0.05324)}$$

$\approx 0.7519$.

Thus the predicted prob. that a company will issue a dividend this year. is 75.2%, and since it is > 50%. the Bayes classifier will predict Yes.